

EMCH 792: Optimal State Estimation
Homework 1
Computer Portion Solutions

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Problem 1: Vector Operations (10 pts)

Create two vectors $\mathbf{v}_1 = [-4, 3, 5]^T$ and $\mathbf{v}_2 = [1, 0, 7]^T$. Find the scalar (inner) product and outer product. Show that $\mathbf{v}_1^T \mathbf{v}_2 = \text{Tr}(\mathbf{v}_1 \mathbf{v}_2^T)$.

MATLAB Code

```

1 v_1 = [-4; 3; 5];
2 v_2 = [1; 0; 7];
3
4 scalar_product = v_1' * v_2
5 outer_product = v_1 * v_2'
6 checkEqual(v_1' * v_2, trace(v_1 * v_2'))

```

MATLAB Output

```

scalar_product =
    31

outer_product =
   -4     0   -28
     3     0    21
     5     0    35

The two inputs are equal

```

Results

Scalar (inner) product:

$$\mathbf{v}_1^T \mathbf{v}_2 = (-4)(1) + (3)(0) + (5)(7) = 31$$

Outer product:

$$\mathbf{v}_1 \mathbf{v}_2^T = \begin{pmatrix} -4 \\ 3 \\ 5 \end{pmatrix} \begin{pmatrix} 1 & 0 & 7 \end{pmatrix} = \begin{pmatrix} -4 & 0 & -28 \\ 3 & 0 & 21 \\ 5 & 0 & 35 \end{pmatrix}$$

Trace identity: $\text{Tr}(\mathbf{v}_1 \mathbf{v}_2^T) = -4 + 0 + 35 = 31 = \mathbf{v}_1^T \mathbf{v}_2 \quad \checkmark$

Problem 2: Matrix Properties (15 pts)

Define:

$$A = \begin{pmatrix} -3 & 5 & 7 \\ 1 & 2 & 4 \\ 6 & 8 & -9 \end{pmatrix}, \quad B = \begin{pmatrix} 8 & 1 & 6 \\ -3 & 5 & 7 \\ 4 & 9 & -2 \end{pmatrix}$$

Problem 2.1: Transpose Identities

Show that $(A^T)^T = A$, $(AB)^T = B^T A^T$, and $(A + B)^T = A^T + B^T$.

MATLAB Code

```
1 A = [-3, 5, 7; 1, 2, 4; 6, 8, -9];  
2 B = [8, 1, 6; -3, 5, 7; 4, 9, -2];  
3  
4 checkEqual((A')', A)  
5 checkEqual((A * B)', B' * A')  
6 checkEqual((A + B)', A' + B')
```

MATLAB Output

```
The two inputs are equal  
The two inputs are equal  
The two inputs are equal
```

Results

All three transpose identities verified:

- $(A^T)^T = A$ ✓
- $(AB)^T = B^T A^T$ ✓
- $(A + B)^T = A^T + B^T$ ✓

Problem 2.2: Rank, Determinant, and Inverse

Find the rank and determinant of A and B . If invertible, find their inverse.

MATLAB Code

```
1 rank_A = rank(A)
2 det_A = det(A)
3 inv_A = inv(A)
4
5 rank_B = rank(B)
6 det_B = det(B)
7 inv_B = inv(B)
```

MATLAB Output

```
rank_A = 3      det_A = 287
rank_B = 3      det_B = -844
```

Results

Both matrices are full rank (rank = 3) with nonzero determinants, so both are invertible.

$$A^{-1} = \begin{pmatrix} -0.1742 & 0.3519 & 0.0209 \\ 0.1150 & -0.0523 & 0.0662 \\ -0.0139 & 0.1882 & -0.0383 \end{pmatrix}$$

$$B^{-1} = \begin{pmatrix} 0.0865 & -0.0664 & 0.0273 \\ -0.0261 & 0.0474 & 0.0877 \\ 0.0557 & 0.0806 & -0.0509 \end{pmatrix}$$

Problem 2.3: Inverse of a Product

Show that $(AB)^{-1} = B^{-1}A^{-1}$.

MATLAB Code

```
1 checkEqual(inv(A * B), inv(B) * inv(A))
```

MATLAB Output

The two inputs are equal

Results

$$(AB)^{-1} = B^{-1}A^{-1} \quad \checkmark$$

Problem 2.4: Eigenvalues and Eigenvectors

Find the eigenvalues and eigenvectors of A .

MATLAB Code

```
1 [V, D] = eig(A)
```

MATLAB Output

```
V =  
    0.4663    -0.8580    -0.6475  
    0.1874     0.4159    -0.5549  
   -0.8646    -0.3016    -0.5224  
  
D =  
  -13.9697         0         0  
         0    -2.9633         0  
         0         0     6.9330
```

Results

The eigenvalues of A are:

$$\lambda_1 \approx -13.970, \quad \lambda_2 \approx -2.963, \quad \lambda_3 \approx 6.933$$

The corresponding eigenvectors (columns of V) are:

$$\mathbf{v}_1 \approx \begin{pmatrix} 0.466 \\ 0.187 \\ -0.865 \end{pmatrix}, \quad \mathbf{v}_2 \approx \begin{pmatrix} -0.858 \\ 0.416 \\ -0.302 \end{pmatrix}, \quad \mathbf{v}_3 \approx \begin{pmatrix} -0.648 \\ -0.555 \\ -0.522 \end{pmatrix}$$

Problem 3: Matrix Inversion Lemma — Example 1.1 (25 pts)

Verify the matrix inversion lemma (Equation 1.45) using Example 1.1 (p. 12). Given:

$$A = \begin{pmatrix} -2 & 1 & 2 \\ -4 & 3 & 2 \\ -5 & 1 & 5 \end{pmatrix}, \quad A' = \begin{pmatrix} -2 & 1 & 2 \\ -4 & 3 & 2 \\ -5 & 1 & 6 \end{pmatrix}$$

where $A' = A + BDC$ with $B = [0; 0; 1]$, $C = [0 \ 0 \ 1]$, $D = 1$.

The matrix inversion lemma states:

$$(A + BD^{-1}C)^{-1} = A^{-1} - A^{-1}B(D + CA^{-1}B)^{-1}CA^{-1}$$

MATLAB Code

```

1 A = [-2 1 2; -4 3 2; -5 1 5];
2 A_inv = inv(A);
3 A_acute = [-2 1 2; -4 3 2; -5 1 6];
4 B = [0; 0; 1];
5 C = [0 0 1];
6 D = 1;
7
8 A_acute_inv_lemma = A_inv - A_inv*B*inv(D + C*A_inv*B)*C*A_inv
9 A_acute_inv_matlab = inv(A_acute)
10 checkEqual(A_acute_inv_lemma, A_acute_inv_matlab)

```

MATLAB Output

```

A_acute_inv_lemma =
    4.0000    -1.0000    -1.0000
    3.5000    -0.5000    -1.0000
    2.7500    -0.7500    -0.5000

```

```

A_acute_inv_matlab =
    4.0000    -1.0000    -1.0000
    3.5000    -0.5000    -1.0000
    2.7500    -0.7500    -0.5000

```

```

The two inputs are equal

```


Results

The matrix inversion lemma yields:

$$(A')^{-1} = A^{-1} - A^{-1}B(D + CA^{-1}B)^{-1}CA^{-1} = \begin{pmatrix} 4 & -1 & -1 \\ 3.5 & -0.5 & -1 \\ 2.75 & -0.75 & -0.5 \end{pmatrix}$$

This matches MATLAB's `inv(A_acute)` exactly. ✓